Amine Maazizi

Research Intern at EPFL | MVA Student, ENS Paris-Saclay | Engineering Student, ENSTA Paris
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Website | GitHub | LinkedIn

Research Interests

Structure-aware representation learning for scientific data. I am interested in machine learning methods that encode inductive biases to improve generalization beyond scaling. My current work studies latent-space perturbation models for spatial proteomics and treatment-response simulation.

Education

ENS Paris-Saclay **2025 – 2026**
Master in Mathematics, Vision and Learning (MVA)
Relevant coursework:Geometric Deep Learning, Geometric Processing Geometric Data Analysis, Representation Learning, Graphs in Machine Learning, Multimodal Explainable AI.

ENSTA Paris, Institut Polytechnique de Paris **2023 – 2026**
Engineering Degree in Computer Science and Artificial Intelligence GPA: **4.05/4.3**, Top 5%
Relevant coursework: Machine Learning, Statistical Learning, Deep Learning, Computer Vision, Language Models, Reinforcement Learning.

CPGE Moulay Youssef, Rabat, Morocco **2021 – 2023**
Preparatory Classes (MPSI / MP)* Top 5% nationally

Research Experience

EPFL, Artificial Intelligence in Molecular Medicine Lab, Lausanne, Switzerland **Apr 2026 – Sep 2026**
Research Intern
Supervisor: Dr. Lukas Klein
Reformulating treatment-response simulation as a latent-space perturbation modeling problem over spatial proteomics representations, aiming to learn structured intervention dynamics that generalize across biological contexts.

Institut Pasteur, Paris, France **Jun 2025 – Dec 2025**
Research Intern and Collaborator
Supervisor: Prof. Giacomo Nardi
Built a geometry-aware shape-matching framework with mechanical priors for morphodynamic analysis, supporting a curvature-driven study of endothelial-to-hematopoietic transition.

NAIST, Cybernetics and Reality Engineering Lab, Remote **Oct 2024 – Dec 2024**
Research Collaborator
Supervisor: Prof. Hideaki Uchiyama
Authored a structured survey of text- and image-conditioned 3D generation methods, clarifying the field's shift from geometric inductive biases toward 2D generative priors.

Selected Research Output

Morphodynamic Study of Hematopoietic Stem Cell Emergence Using Shape Matching with Mechanical Constraints
G. Nardi, **A. Maazizi**, et al.
Manuscript in preparation

Software and Projects

not-MIWAE PyTorch GitHub / PyPI
PyTorch package implementing MIWAE, not-MIWAE, supMIWAE, and supervised MNAR extensions in a unified missing-data modeling framework.

Geometry-Aware Shape Matching Private Code

Geometry-aware mesh-matching framework with configurable mechanical priors for constrained shape comparison and morphodynamic analysis.

SAGALang Project

Experimental programming-language project exploring syntax and implementation design.

Awards and Honors

Finalist, Best Research Project Award, ENSTA Paris **2025**

Moroccan Excellence Scholarship, Ministry of Higher Education **2023**

Technical Skills

Programming: Python, C, C++
Machine Learning: PyTorch, JAX, TensorFlow, PyTorch Geometric
Scientific Computing: NumPy, SciPy
Infrastructure: Linux, Git, HPC (SLURM)

Languages

Arabic (Native) | French (Bilingual) | English (Fluent) | Spanish (Elementary)